EDO STATE GOVERNMENT

Integrated Land Administration & Revenue Automation Platform (LERA)

A Comprehensive Proposal for Digital Transformation of Property-Based Revenue Collection, Monitoring, and Fiscal Governance Across Edo State

Prepared for: Edo State Government

Edo State Internal Revenue Service (EIRS)

Ministry of Finance & Economic Development

Local Government Areas & Revenue Agencies

Document Classification: Official Proposal — Confidential

Version: 1.0 | **Date:** June 2026

Executive Summary

Edo State, like many sub-national governments in Nigeria, faces a structural challenge that is as much administrative as it is fiscal: the gap between the state's revenue potential and the revenue actually collected, verified, and accounted for. Property-based revenues — including Land Use Charge (LUC), tenement rates, development levies, and related statutory charges — represent one of the most sustainable, equitable, and scalable sources of internally generated revenue (IGR). Yet without a unified digital platform, these revenues remain fragmented across manual registers, inconsistent valuation practices, delayed bill issuance, weak collection follow-up, and limited visibility for policymakers.

This proposal presents the **Land Administration & Revenue Automation (LERA) Platform** — a proven, enterprise-grade solution designed to digitize the entire property-to-payment lifecycle for state and local government revenue authorities. LERA is not merely a billing software package. It is an integrated governance system that connects property enumeration, taxpayer registration, automated assessment, multi-level approval workflows, multi-channel payment collection, real-time dashboards, reconciliation, settlement reporting, and audit trails into a single coherent architecture.

> **Strategic value for Edo State:** LERA enables the State Government to move from reactive, paper-driven revenue administration to proactive, data-driven revenue mobilization — with measurable improvements in collection rates, transparency, taxpayer compliance, and inter-agency coordination across all Local Government Areas (LGAs) and revenue-mandated agencies.

Key Outcomes Anticipated

Dimension	Current Challenge	LERA-Enabled Outcome
Revenue Visibility	**Delayed, fragmented reporting**	**Real-time statewide dashboard by LGA, zone, and ward**
Property Registry	**Incomplete or duplicated records**	**Centralized GIS-linked property database with sync capability**

Bill Issuance	**Manual, inconsistent assessments**	**Rules-based automated billing with approval controls**
Collection	**Cash-heavy, low traceability**	**Online payments, bank channels, QR codes, digital receipts**
Accountability	**Limited audit trails**	**Role-based access, activity logs, reconciliation modules**
Policy Decisions	**Intuition-driven planning**	**Analytics on arrears, compliance, zone performance, trends**

Scope of This Document

This proposal outlines the policy rationale for adopting LERA in Edo State, describes the platform's functional modules in detail, explains how the system supports revenue monitoring and generation, presents a phased implementation roadmap, and articulates the governance, security, and sustainability considerations necessary for long-term success. The document is intended for review by the Executive Governor's office, the Honourable Commissioner for Finance, EIRS leadership, LGA chairmen, and relevant legislative oversight committees.

LERA has been engineered for multi-agency deployment with sector-based access control, allowing Edo State to operate a single statewide platform while preserving appropriate data boundaries for LGAs, special assessment zones, and delegated revenue agencies — all under unified executive oversight.

Table of Contents

1. Executive Summary
2. Introduction & Policy Context
3. Background: Revenue Administration in Sub-National Governments
4. Problem Statement & Gap Analysis
5. Proposed Solution: The LERA Platform
6. System Architecture Overview
7. Module 1 — Property & Taxpayer Registry
8. Module 2 — GIS Synchronization & Data Quality
9. Module 3 — Billing Engine & Assessment Rules
10. Module 4 — Workflow, Approvals & Governance Controls
11. Module 5 — Bill Printing, Delivery & Citizen Engagement
12. Module 6 — Payment Collection & Digital Channels
13. Module 7 — Receipts, Reconciliation & Settlement
14. Module 8 — Dashboard, Analytics & Executive Monitoring
15. Module 9 — Reporting & Regulatory Compliance
16. Module 10 — Objections, Appeals & Taxpayer Rights
17. Module 11 — Security, RBAC & Audit
18. Module 12 — Multi-Agency & LGA Configuration
19. Revenue Generation Strategy for Edo State
20. Monitoring & Performance Management Framework
21. Implementation Roadmap & Phasing
22. Change Management, Training & Capacity Building
23. Risk Management & Mitigation
24. Conclusion & Call to Action
25. Appendix A — Technical Specifications

26. Appendix B — User Roles & Permissions Matrix
27. Appendix C — Payment Integration Overview

1. Introduction & Policy Context

Edo State Government has demonstrated consistent commitment to economic development, infrastructure investment, and public service delivery. Sustainable funding for these priorities depends fundamentally on the state's capacity to mobilize domestic revenue efficiently, fairly, and transparently. Federal allocations, while important, are insufficient and unpredictable as the sole basis for developmental expenditure. Internally Generated Revenue (IGR) — particularly property-linked charges rooted in land use, occupancy, and development — must therefore occupy a central place in Edo State's fiscal strategy.

The Land Use Act (1978) and subsequent state-level legislation vest land administration and related revenue powers in state and local authorities. In practice, effective collection requires more than legal authority: it requires accurate property data, defensible valuations, timely billing, accessible payment channels, robust enforcement mechanisms, and executive dashboards that allow the Governor, Commissioner for Finance, and EIRS leadership to monitor performance in real time.

LERA addresses this requirement holistically. Developed as a full-stack enterprise platform comprising a secure REST API backend and a modern web-based administrative portal, LERA has been designed for deployment across multiple agencies and LGAs within a single state ecosystem. The platform supports the complete revenue chain from property discovery to payment confirmation, receipt issuance, bank reconciliation, and inter-agency settlement.

2. Background: Revenue Administration in Sub-National Governments

Across Nigeria, sub-national revenue authorities face recurring operational constraints that limit IGR performance:

- **Data fragmentation:** Property records may exist in GIS databases, paper field enumeration forms, LGA registers, and legacy spreadsheets — often without a golden source of truth.
- **Manual assessment:** Valuation and charge computation depend on individual officer judgment, producing inconsistency and disputes.

- **Weak workflow controls:** Bills may be issued without adequate review, authorization, or segregation of duties — creating fiscal and reputational risk.
- **Limited payment options:** Taxpayers face inconvenience when only cash or in-person payment is available, depressing compliance.
- **Reconciliation gaps:** Bank deposits, payment gateway settlements, and bill records frequently fail to align, obscuring true collection performance.
- **Delayed reporting:** Executives receive revenue data weeks or months after transactions occur, preventing agile policy intervention.

Digital revenue platforms have reversed these trends in jurisdictions that have invested in integrated systems. Automation reduces leakage. Workflow enforcement reduces fraud. Online payment increases compliance. Dashboards enable target-setting and accountability at LGA and ward levels. LERA embodies these principles in a solution already architected for Nigerian revenue administration practice.

3. Problem Statement & Gap Analysis

3.1 Revenue Leakage & Under-Assessment

Without a comprehensive property register linked to automated assessment rules, significant portions of the taxable base remain outside the billing net. New developments, change of use, and occupancy modifications are often detected late — if at all. Minimum charge floors, depreciation schedules, and property classification rules are inconsistently applied.

3.2 Collection Inefficiency

Physical bill distribution is costly and slow. Taxpayers who are willing to pay may be unable to do so conveniently. Arrears accumulate not solely from unwillingness to pay but from friction in the payment process and uncertainty about amounts due.

3.3 Weak Executive Oversight

State leadership requires answers to straightforward questions: How much was billed this year? How much was collected? Which LGA is underperforming? What is the arrears position by zone? Without integrated data, these questions require manual compilation — often

producing conflicting figures from different departments.

3.4 Compliance & Trust Deficit

Taxpayers who cannot verify their assessments, obtain receipts promptly, or lodge formal objections through a transparent process are less likely to comply voluntarily. A fair, auditable system strengthens the social contract between government and citizens.

4. Proposed Solution: The LERA Platform

LERA is proposed as Edo State's statewide integrated platform for property-based revenue administration. The solution comprises:

- **LERA Backend API** — Laravel-based enterprise service handling business logic, integrations, queues, PDF generation, exports, and webhooks.
- **LERA Admin Portal** — Angular-based web application for revenue officers, approvers, LGA administrators, and executives.
- **Integration Layer** — Connections to payment gateways (RoutePay, eTranzact, Odoo payment portal), GIS enumeration databases, and email notification services.
- **Document Services** — Automated generation of demand notices, batch bill PDFs with payment QR codes, payment receipts, and regulatory reports.

> **Deployment model:** A single statewide instance with sector-scoped access ensures Edo State maintains unified data standards and executive visibility, while LGAs and delegated agencies operate within their authorized geographic and functional boundaries.

The platform is cloud-ready, queue-driven for heavy operations (billing runs, batch PDF generation, large exports), and designed for horizontal scaling as transaction volumes grow.

5. System Architecture Overview

LERA follows a modern three-tier architecture optimized for security, maintainability, and integration:

Layer	Technology	Responsibility
Presentation	Angular SPA, PrimeNG, ApexCharts	User interface, dashboards, role-based navigation, exports download
Application	Laravel 11 REST API, JWT Auth	Business rules, workflows, permissions, job queues, PDF/Excel generation
Data	MySQL / PostgreSQL	Transactional data, property registry, bills, payments, audit logs
Integration	HTTP webhooks, OAuth, GIS sync	Payment gateways, enumeration databases, notification services

5.1 Core Design Principles

- **API-first:** All functionality exposed via documented REST endpoints, enabling future mobile apps and citizen portals.
- **Asynchronous processing:** Heavy tasks (annual billing, batch PDFs, bulk exports) run in background queues without blocking users.
- **Sector multi-tenancy:** Users are scoped to authorized LGAs/agencies; data queries automatically enforce boundaries.
- **Auditability:** Critical actions (deletions, payment posting, workflow transitions) are logged.
- **Configurable rules:** Charge rates, depreciation, minimum LUC, relief settings, and agency payment credentials are administratively configurable.

6. Module 1 — Property & Taxpayer Registry

The foundation of any property tax system is an accurate, searchable register of land parcels, buildings, owners, and occupancy characteristics. LERA's property module provides:

6.1 Owner Management

- Registration and maintenance of property owner records with KGTIN (tax identification), contact details, and correspondence addresses.
- Search and lookup by identification number, name, or linked properties.
- Validation of contact data for billing and notification purposes.

6.2 Property Registry

- Comprehensive property profiles: building code, address, ward, zone, LGA, property use/type, classification, assessed value indicators, coordinates, and imagery.
- Paginated statewide listing with sector-based filtering for LGA officers.
- Advanced keyword search across property attributes.
- Controlled editing of property fields with business rule enforcement (e.g., deletion blocked when paid bills exist).

6.3 Exempted Properties

- Dedicated register for properties granted full or partial exemption (government facilities, places of worship where applicable by policy, etc.).
- Dashboard visibility of exempt inventory for policy review.

6.4 Property Exceptions

- Queue of properties failing validation or sync rules (missing classification, incomplete enumeration data, duplicate building codes).
- Exportable exception reports for field remediation campaigns.

Revenue impact: Every bill issued in LERA is anchored to a verified property record, eliminating "ghost bills" and ensuring collection efforts target real assets.

7. Module 2 — GIS Synchronization & Data Quality

Edo State's property enumeration initiatives and GIS investments produce valuable data that must flow into the revenue system automatically rather than through manual re-entry.

- **LGA-level synchronization:** Pull building records from enumeration databases or remote GIS APIs into LERA's property registry.
- **Sync reporting:** Historical logs of synchronization jobs with success/failure metrics.
- **Force sync:** Re-pull individual properties when field corrections are made at source.
- **Bulk re-synchronization:** Queue large remediation batches after enumeration drives.
- **Street name propagation:** Updates street addressing on properties and linked bills when source data changes.

This module ensures the revenue database stays current as Edo State's built environment evolves — critical for capturing new developments in the tax base promptly.

8. Module 3 — Billing Engine & Assessment Rules

LERA's billing engine automates the computationally intensive process of annual (or periodic) Land Use Charge assessment across thousands or millions of properties.

8.1 Master Data Configuration

Before billing runs, administrators configure the rule framework:

- Zones, wards, and geographic hierarchies
- Property classifications (residential, commercial, industrial, mixed-use, etc.)

- Property Assessment Values (PAV) by type and sector
- Charge rates and occupancy multipliers
- Depreciation schedules
- Minimum LUC floors
- Relief and incentive settings
- Area offices and distribution codes

8.2 Bill Processing

- **Preview:** Officers preview eligible properties for a billing year before execution.
- **Async processing:** Billing runs execute as background jobs, computing charges from PAV × rate × depreciation with minimum LUC enforcement.
- **Duplicate prevention:** System skips properties already billed for the target year.
- **Exception logging:** Properties without matching PAV codes are flagged rather than silently skipped.
- **Rollback:** Authorized rollback of unapproved bills for a year when policy corrections require re-processing.

8.3 Bill Lifecycle States

Status	Stage	Description
0	Pending	Newly generated, awaiting review
1	Reviewed	Initial quality check completed
2	Verified	Technical/financial verification
3	Authorized	Management authorization
4	Approved	Final approval — eligible for printing and payment
5	Returned	Sent back for correction with documented reason

9. Module 4 — Workflow, Approvals & Governance Controls

Fiscal governance requires that no bill reaches a taxpayer without passing through defined checkpoints. LERA enforces segregation of duties through a multi-stage workflow:

- **Single bill actions:** Review, verify, authorize, approve, or return individual bills with audit trail.
- **Bulk actions:** Process batches of bills at each workflow stage for operational efficiency.
- **Bill editing:** Authorized officers may adjust assessed values, rates, and amounts prior to final approval.
- **Special interest bills:** Parallel workflow track for high-priority or sensitive assessments requiring enhanced scrutiny.
- **Returned bills queue:** Dedicated views for bills sent back for correction, with search and filtering.
- **Deletion controls:** Bills with recorded payments cannot be deleted; deletions are activity-logged.

For Edo State, this workflow maps directly to EIRS internal control standards and can be aligned with state audit requirements. Each role — Biller, Reviewer, Verifier, Authorizer, Approver — receives only the permissions necessary for their function.

10. Module 5 — Bill Printing, Delivery & Citizen Engagement

10.1 Print Selection & Batch Generation

- Filter approved bills by LGA, zone, ward, distribution code, street name, and property class.
- Initiate print batches that mark bills as printed and assign batch codes.

- Background generation of consolidated batch PDFs with agency-branded demand notice templates.
- Regeneration capability when templates or QR payment links are updated.

10.2 Demand Notice Features

- Professional PDF layout with property details, assessment number, charge breakdown, balance brought forward, and total payable.
- Payment QR codes linking to online payment portals (configurable per agency).
- Proof of delivery (POD) section for field collection agents.
- Agency-specific branding (background templates per LGA/agency).

10.3 Delivery Tracking

- Update delivery status when bills are physically served to taxpayers.
- Delivery metrics on executive dashboard for monitoring field operations.

Citizen benefit: Clear, standardized demand notices with scannable QR codes reduce confusion and enable immediate payment — a proven driver of compliance.

11. Module 6 — Payment Collection & Digital Channels

Revenue realization depends on making payment as easy as possible while maintaining full traceability. LERA supports multiple collection channels suited to Edo State's diverse taxpayer base:

11.1 Online Payment Gateway (RoutePay)

- Hosted checkout initiated from the admin portal or taxpayer-facing payment links.
- Per-agency OAuth credentials and payment gateway configuration.
- Webhook-based settlement confirmation with signature verification and idempotent payment logging.
- Automatic bill status update (paid / partly paid) upon confirmed payment.

11.2 QR Code Payment on Bills

- Each printed demand notice embeds a QR code encoding the agency payment URL.
- Taxpayers scan with any smartphone to reach the payment page instantly.
- Configurable URL templates supporting assessment number, building code, amount, and year parameters.

11.3 Bank / Switch Integration (eTranzact)

- Pre-payment validation endpoint returning taxpayer name, amount due (including arrears), and agency details.
- Payment notification webhook with IP whitelist security for bank-posted transactions.
- Balance brought forward automatically included in validation totals.

11.4 Manual Receipt Workflow

- Counter payments recorded by revenue officers with pending approval status.
- Supervisor approve/discard workflow before payment posts to bill.

- Digital receipt generation upon approval.

11.5 Self-Service Payment

- Public bill lookup by assessment number.
- Payment completion page for online transactions.
- Automated receipt email with PDF attachment after successful payment.

Channel	Taxpayer Experience	Government Benefit
Online (RoutePay/Odoo)	Pay anytime from phone or computer	Instant posting, digital audit trail
QR on bill	Scan and pay in seconds	Higher conversion from bill to payment
Bank/USSD (eTranzact)	Pay at bank or via mobile	Reaches unbanked/underbanked segments
Counter (manual receipt)	Pay at EIRS/LGA office	Controlled cash handling with approval

12. Module 7 — Receipts, Reconciliation & Settlement

12.1 Digital Receipts

- Branded PDF receipts with payment details, revenue code, assessment reference, and verification QR code.
- Public receipt verification via receipt number lookup.
- Downloadable proof of payment for taxpayers.

12.2 Bank Reconciliation

- Upload bank/payment provider settlement files (Excel format).
- Automatic matching against payment logs with exception reporting.

- Reconciliation history with UUID tracking and confirmation/lock for closed periods.
- Unreconciled transaction reports for investigation.
- Re-query and manual reconciliation tools for disputed items.

12.3 Settlement Reporting

- Configurable revenue split percentages (state agency, LGA, partner shares).
- Settlement reports by agency and date range for treasury disbursement.
- Exportable settlement data for accounting systems.

Fiscal integrity: Reconciliation ensures that every Naira collected is accounted for before it reaches the treasury — a critical safeguard for Edo State's public financial management framework.

13. Module 8 — Dashboard, Analytics & Executive Monitoring

LERA transforms raw transaction data into actionable intelligence for policymakers and revenue managers.

13.1 Statewide Executive Dashboard

- Total properties registered statewide
- Approved, pending, and objected bill counts
- Total amount billed vs. total amount collected (₦)
- Bills printed and delivered counts
- **Per-agency breakdown tab:** Same KPIs for each LGA/agency — enabling comparative performance review

13.2 Visual Analytics

- Monthly payment trend charts (area/line)
- Bill distribution by geographic zone (donut/bar)

- Bill and payment comparison by LGA
- Property distribution maps by zone and LGA
- Year-over-year filtering for trend analysis

13.3 LGA Chair Dashboard

- Dedicated view scoped to the chairperson's LGA
- Outstanding bills list and payment history for local oversight
- Localized charts without exposure to other LGAs' data

> **Governor's Office use case:** Weekly review of collection rate by LGA, identification of underperforming zones, and data-driven deployment of enforcement and sensitization resources.

14. Module 9 — Reporting & Regulatory Compliance

LERA includes a comprehensive reporting suite for operational, financial, and compliance purposes:

Report	Purpose	Export
Customer Statement	Payment history by property/building code	PDF, Excel
Payment Report	Collections by LGA, zone, ward, date range	Excel
Workflow Report	Bill pipeline metrics by approval stage	Excel
Performance Report	Staff productivity and processing volumes	Excel
Delivery Report	Bill serving status by area	Excel
Print Report	Printed bill batches and volumes	Excel
Settlement Report	Revenue distribution by agency	Excel
Reconciliation Report	Matched vs. unmatched transactions	Excel
Property Exception Export	Data quality remediation list	Excel
Bill Exports (async)	Outstanding, paid, pending, returned, partly-paid	Excel (queued)

Large exports run asynchronously via background jobs with status polling and download links — ensuring the system remains responsive even when generating reports spanning millions of records.

15. Module 10 — Objections, Appeals & Taxpayer Rights

A fair revenue system must provide structured mechanisms for taxpayers to challenge assessments. LERA's objection module supports:

- Formal objection submission with reason, relief selections, and document attachments.
- Multi-stage review: Request → Verification → Authorization (revised LUC determination) → Approval.
- Upon approval: archival of original bill and issuance of revised assessment.
- Email notifications at status transitions.
- Searchable objection queues and dedicated objected-bills views.

This protects Edo State from arbitrary assessment practices while giving EIRS a defensible, documented process for handling disputes — reducing litigation risk and building taxpayer confidence.

16. Module 11 — Security, RBAC & Audit

16.1 Authentication

- JWT-based API authentication with token expiration and logout.
- Forced password change for default credentials.
- Typed user login supporting role-specific portals.

16.2 Role-Based Access Control

- Granular permission codes (50+ permissions) controlling menu visibility and API access.
- Roles: Administrator, Biller, Reviewer, Verifier, Authorizer, Approver, LGA Chair, and custom roles.
- Permission assignment interface for security administrators.

16.3 Sector Scoping

- Users assigned to one or more LGAs/agencies via sector field.
- Automatic query filtering — officers cannot access data outside their sector.
- 403 enforcement on unauthorized LGA access attempts.

16.4 Audit Trail

- Activity logging for bill deletions, property deletions, receipt posting, and critical state changes.
- Webhook idempotency and timestamp tolerance for payment integrity.

17. Module 12 — Multi-Agency & LGA Configuration

Edo State comprises multiple LGAs and potentially delegated revenue agencies. LERA supports this structure through:

- **Agency registry:** Create and manage LGAs with contact details and operational parameters.
- **Per-agency payment gateway:** Independent RoutePay credentials, return URLs, and webhook endpoints.
- **Agency-branded documents:** Custom PDF backgrounds and logos per agency.
- **Sector assignment:** Map users to agencies for scoped operations.
- **Distribution codes & zones:** Geographic subdivisions within agencies for targeted operations.

This architecture allows Edo State to implement a "single platform, multiple operators" model — the gold standard for statewide revenue systems.

18. Revenue Generation Strategy for Edo State

Adopting LERA is not simply an IT procurement exercise — it is a revenue mobilization strategy. The following mechanisms translate platform capabilities into measurable fiscal outcomes:

18.1 Broadening the Tax Base

GIS synchronization and property exception remediation ensure that newly developed properties, commercial conversions, and previously unenumerated buildings enter the billing cycle. Industry experience suggests that comprehensive digitization of property registers can expand the effective tax base by 15–40% within the first three billing cycles, depending on starting data quality.

18.2 Reducing Arrears Through Payment Convenience

QR-enabled demand notices, online payment portals, and bank channel integration remove friction that causes willing taxpayers to delay payment. Balance brought forward calculations on bills make arrears visible and payable in a single transaction.

18.3 Accelerating Billing Cycles

Automated assessment replaces weeks of manual computation with queue-driven processing that can generate hundreds of thousands of bills in hours — enabling Edo State to issue demand notices at the start of each fiscal year rather than mid-year.

18.4 Targeted Enforcement

Outstanding bill reports filterable by LGA, zone, ward, and street enable focused enforcement campaigns. Dashboard metrics identify LGAs with high bill-to-collection ratios for intervention.

18.5 Reducing Leakage

Workflow controls prevent unauthorized bill issuance. Manual receipts require supervisor approval. Reconciliation matches collections to bank deposits. Settlement reports ensure correct revenue sharing. Together, these controls address the primary sources of revenue leakage in manual systems.

18.6 Projected Revenue Impact (Illustrative)

Scenario	Assumption	Illustrative Annual Impact
Base expansion	20% more properties billed via GIS sync	+20% on property tax component of IGR
Collection rate improvement	Compliance rises from 45% to 60%	+33% on amounts actually collected
Leakage reduction	5% of collections previously unaccounted	Direct recovery to treasury
Arrears recovery	10% of BBF collected via QR/online	One-time and recurring arrears inflow

Note: Actual projections require baseline data from Edo State EIRS. LERA's reporting module will generate the baseline within 90 days of property data migration.

19. Monitoring & Performance Management Framework

LERA enables Edo State to institute a formal revenue performance management framework:

19.1 Key Performance Indicators (KPIs)

- **Billing coverage ratio:** Properties billed ÷ total registered properties
- **Collection efficiency ratio:** Amount collected ÷ amount billed
- **Arrears ratio:** Outstanding balance ÷ total assessed

- **Digital payment ratio:** Online/bank payments ÷ total payments
- **Bill delivery rate:** Bills delivered ÷ bills printed
- **Objection resolution time:** Average days from submission to decision
- **Reconciliation match rate:** Matched transactions ÷ total bank entries
- **LGA comparative ranking:** Collection performance index by agency

19.2 Reporting Cadence

Audience	Report	Frequency
Governor / Commissioner	Statewide dashboard KPIs	Weekly
EIRS Management	Collection, workflow, performance reports	Daily / Weekly
LGA Chairmen	LGA-scoped dashboard	Weekly
Treasury / Accountant-General	Settlement & reconciliation reports	Monthly
Audit / Legislature	Annual billing & collection summary	Quarterly / Annual

19.3 Accountability Mechanisms

- Performance reports attribute bill processing and payment recording to individual officers.
- LGA comparative dashboards create constructive competition among local governments.
- Workflow reports expose bottlenecks (e.g., bills stuck at verification stage).

20. Implementation Roadmap & Phasing

A phased rollout minimizes disruption and allows Edo State to demonstrate quick wins while building toward full statewide coverage.

Phase	Duration	Activities	Deliverables
Phase 1: Foundation	Months 1–2	Infrastructure setup, agency configuration, master data load, user provisioning, training of administrators	Production environment, configured agencies, admin users trained
Phase 2: Property Migration	Months 2–4	GIS sync for pilot LGAs, exception remediation, owner data cleanup	Validated property register for pilot areas
Phase 3: Pilot Billing	Month 4–5	Billing run for pilot LGAs, full workflow test, print batch, payment channel test	First digital bills issued and collected
Phase 4: Statewide Rollout	Months 5–8	Remaining LGAs onboarded, statewide billing, public sensitization campaign	All LGAs live on LERA
Phase 5: Optimization	Months 8–12	Reconciliation automation, objection module activation, citizen self-service portal, mobile roadmap	Full platform capability utilized

21. Change Management, Training & Capacity Building

Technology alone does not transform revenue administration — people and processes must evolve alongside the platform.

- **Executive briefing:** Half-day session for Governor's economic team, Commissioner, and EIRS leadership on dashboard usage and KPI interpretation.

- **Train-the-trainer:** EIRS IT and operations staff certified to train LGA users.
- **Role-based training:** Separate curricula for Billers, Reviewers, Approvers, Cashiers, and LGA Chairs.
- **Standard Operating Procedures:** Documented workflows aligned with LERA system states.
- **Helpdesk:** Tier-1 support at EIRS; Tier-2 technical support from implementation partner.
- **Public sensitization:** Taxpayer education on QR payment, online portals, and objection rights.

22. Risk Management & Mitigation

Risk	Mitigation
Poor source data quality	Property exception module, phased migration, field validation campaigns
User resistance to change	Executive sponsorship, training, parallel run period, quick wins in pilot LGAs
Payment gateway downtime	Multiple channels (online, bank, counter); manual receipt fallback
Cybersecurity threats	JWT auth, RBAC, HTTPS, webhook signatures, activity logging, regular security updates
Infrastructure failure	Cloud hosting with backups, queue retry logic, disaster recovery plan
Legal challenge to assessments	Workflow audit trail, objection module, documented valuation rules

23. Conclusion & Call to Action

Edo State stands at a decisive moment in its fiscal history. The state's development ambitions — in education, healthcare, infrastructure, security, and job creation — require a revenue base that is broad, compliant, and professionally administered. Continuing with

fragmented manual processes will perpetuate under-collection, weak oversight, and taxpayer distrust. Adopting LERA offers a proven, comprehensive path to modern revenue administration.

The LERA platform addresses every stage of the property revenue lifecycle: from property discovery to payment confirmation, from executive dashboards to bank reconciliation. It is designed for the Nigerian governance context — multi-LGA, multi-channel, multi-role — and ready for statewide deployment.

> **We respectfully recommend that the Edo State Government:**
>
> 1. Approve the adoption of the LERA platform as the state's integrated property revenue system.
> 2. Mandate EIRS to lead implementation in coordination with LGA chairmen and the Ministry of Finance.
> 3. Approve a phased rollout beginning with a two-LGA pilot within 60 days.
> 4. Allocate resources for data migration, training, and public sensitization.
> 5. Establish a steering committee with quarterly review of KPIs from the LERA executive dashboard.

We remain available to conduct a live platform demonstration, facilitate a technical discovery session with EIRS ICT leadership, and support the preparation of cabinet memoranda and budget proposals for implementation funding.

Respectfully submitted,

_____

Project Implementation Team

Land Administration & Revenue Automation (LERA)

June 22, 2026

Appendix A — Technical Specifications

Component	Specification
Backend Framework	Laravel 11 (PHP 8.2+)
Frontend Framework	Angular (TypeScript), PrimeNG, Bootstrap
Authentication	JWT (tymon/jwt-auth)
Database	MySQL 8+ / PostgreSQL 14+
PDF Generation	DomPDF (barryvdh/laravel-dompdf), FPDI merge
Excel Export	Maatwebsite Excel 3.x
Job Queue	Laravel Queue (database/Redis), Laravel Horizon
Payment Integrations	RoutePay (OAuth + webhook), eTranzact, Odoo portal
QR Generation	QuickChart.io API (embedded on PDF bills)
Email	Laravel Mail (queued receipt notifications)
API Style	RESTful JSON API
Hosting	Linux server (Ubuntu), Nginx/Apache, SSL/TLS required
Minimum Server (pilot)	2 vCPU, 4GB RAM, 70GB SSD
Recommended (statewide)	4+ vCPU, 8GB+ RAM, managed database, Redis for queues

Appendix B — User Roles & Permissions Matrix (Summary)

Role	Primary Functions	Key Permissions
System Administrator	Platform configuration, user management, agency setup	All permissions

Biller	Process bills, edit assessments	PROCESS_BILLING, bill queues
Reviewer	First-level bill review	Workflow status 0→1
Verifier	Technical verification	Workflow status 1→2
Authorizer	Management authorization	Workflow status 2→3
Approver	Final bill approval	Workflow status 3→4, PRINT_BILLS
Cashier / Receipt Officer	Manual receipt creation	NEW_RECEIPT, MANAGE_RECEIPT
Report Analyst	Generate and export reports	REPORT_* permissions
Reconciliation Officer	Bank reconciliation processing	RECONCILIATION_*
LGA Chair	LGA-scoped oversight dashboard	LGA_CHAIR_DASHBOARD

Appendix C — Payment Integration Overview

C.1 RoutePay Hosted Checkout Flow

1. Officer or taxpayer initiates payment from LERA portal.
2. System registers transaction with RoutePay and returns redirect URL.
3. Taxpayer completes payment on RoutePay hosted page.
4. RoutePay sends webhook notification to LERA API.
5. LERA verifies signature, logs payment, updates bill status, queues receipt email.
6. Taxpayer redirected to payment completion page.

C.2 Bill QR Payment Flow

1. Approved bill printed with QR code encoding payment URL.
2. Taxpayer scans QR with smartphone camera.

3. Browser opens agency payment portal (e.g., Odoo) with assessment number pre-filled.
4. Payment completed on portal; confirmation flows back via webhook or manual reconciliation.

C.3 eTranzact Bank Channel Flow

1. Taxpayer presents assessment number at bank or via USSD.
2. Bank calls LERA validation API — receives name, amount, agency.
3. Upon payment, bank sends notification to LERA.
4. LERA posts payment and generates receipt.

Appendix D — Glossary

Term	Definition
LUC	Land Use Charge — property-based levy on land and buildings
IGR	Internally Generated Revenue
EIRS	Edo State Internal Revenue Service
LGA	Local Government Area
PAV	Property Assessment Value — unit valuation for charge computation
BBF	Balance Brought Forward — arrears from prior years
KGTIN	Kogi/State Tax Identification Number for taxpayers
Assessment No.	Unique identifier for a bill instance
Building Code	Unique identifier for a property in the registry
LERA	Land Administration & Revenue Automation Platform

This document is proprietary and prepared exclusively for the Edo State Government. Unauthorized reproduction or distribution is prohibited. © 2026 LERA Platform.